For Examiner's Use

6 **(a)** An alternating voltage of period 10 ms is being applied directly across a resistor of 5.0 Ω in a circuit. The variation with time *t* of voltage *V* is shown in Fig. 6.1.

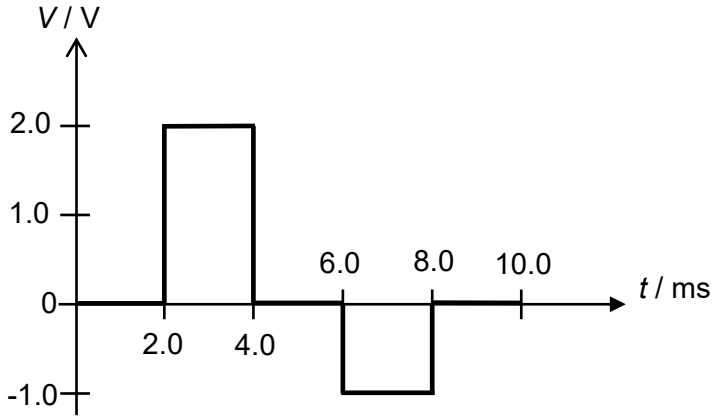

Fig. 6.1

Calculate the steady voltage passing through the same resistor that would produce an identical heating effect.

voltage = ... V [2]

(b) Explain why it is necessary to use high voltages for the efficient transmission of electrical energy.

.....

.....

.....[2]

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(c) Another sinusoidal voltage input of 6.5 mV r.m.s. and 50 Hz is now connected to the primary coil of a transformer as shown in Fig. 6.2. The transformer is assumed to be ideal and its turns ratio, $\frac{N_s}{N_p}$ is 71. The secondary coil is connected to a resistor *R*. An average power of 0.040 W is produced in resistor *R*.

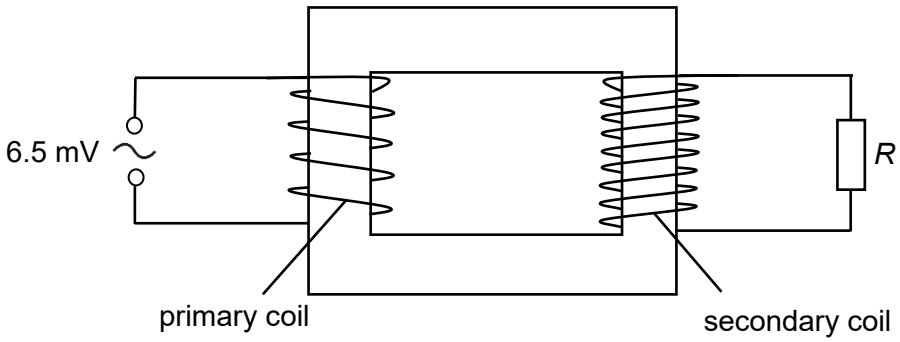

Fig. 6.2

(i) Calculate the r.m.s output voltage supplied to resistor *R*.

r.m.s. voltage = ……. V [1]

(ii) In Fig. 6.3, sketch the variation with time *t* of the power *P* dissipated in the resistor *R*. Label all values on the axes. [2]

Fig. 6.3

For Examiner's Use

(iii) An ideal diode is now connected to the secondary coil with resistor *R* as shown in Fig. 6.4.

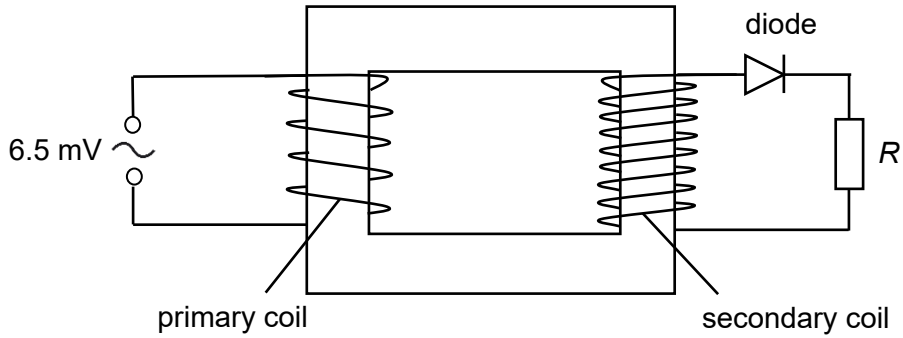

Fig. 6.4

Describe the variation with time of the

1. current flow through resistor *R*, and

.....

.....

.....[2]

2. voltage across resistor *R*.

.....

.....[1]